

Ajit J Gupta

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🐙 [GitHub](#) | [LinkedIn](#) | [LeetCode](#)

EDUCATION

BMS College Of Engineering, Bangalore

B.tech in EEE

2020 – 2024

CGPA: 9.2/10

WORK EXPERIENCE

Hinge Health, Bangalore

July 2024 - Present

Data Engineer

- Designed and **optimized PySpark** pipelines to transform large-scale event data from raw to curated layers, enabling real-time marketing automation via Iterable.
- Replaced a 10-minute batch job (~\$600/month cost) with an **optimized PySpark streaming** solution, **reducing** latency by 90% and saving 60% in monthly costs.
- Built and owned a **Kafka-based streaming** service to consume, enrich, and publish high-volume user events for downstream consumers.
- Orchestrated batch and **streaming** pipelines using **Apache Airflow**, implementing **SLA** monitoring, retries, and dependency management

Hinge Health, Bangalore

Aug 2023 - June 2024

Software Development Engineer Intern

- Developed and maintained a **FastAPI**-based computer vision microservice that processed 10,000+ exercise videos weekly, enabling near real-time batch tracking across **AWS ECS** environments.
- Integrated third-party **CV engine (wrnch)** for pose estimation and built a scalable parallel worker architecture, **reducing** video processing time through **optimized** batch sizing.
- Architected scalable **AWS** infrastructure using **CloudFormation**, supporting multi-version **CV engine** deployment with **auto-scaling ECS** clusters handling up to 500 concurrent jobs during peak loads

Samsung, Bangalore

Dec 2022 - June 2023

R&D Intern

- Built a **machine learning**-powered backend system to predict optimal WiFi access points, solving network handoff and performance issues.
- Exposed ML models via **Flask** REST APIs, enabling seamless integration with client applications.
- Designed backend data pipelines to clean, transform, and store real-time WiFi scan data in **MongoDB** for training.

PROJECT WORK

- AI Agent Verse:** Implemented distributed multi-agent architecture using Google's A2A protocol, deploying 8+ independent agent processes with domain-specific capabilities, enabling seamless inter-agent communication and plug-and-play extensibility.

SKILLS

Languages: C++, Python, TypeScript, SQL, Terraform

Frameworks: Nest.js, Flask, FastAPI, Pytest

Database: MySQL, Postgres, MongoDB

Technologies & Tools: AWS, Databricks, PySpark, Airflow, Kafka, Docker

AWARDS AND CERTIFICATES

- Databricks Certified Data Engineer Associate Certification
- Received Engineering award for my contributions in Hinge Health
- Made open-source contributions to Apache Airflow with 4 pull requests successfully merged
- Won national level Hack-A-Thon organized by PWskills (Physics Wallah) and others.